

# DEFTECH

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Start-ups like ideaForge, IG Drones, NewSpace Research were pivotal during Operation Sindoor, contributing drones, AI-powered surveillance systems and imaging

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India's defence production reached ₹1.27 lakh crore in FY24. Exports soared to ₹23,622 crore in FY25, a 34-fold increase since FY14

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Despite the momentum, the start-ups face steep entry barriers. Inconsistent demand, long procurement cycles, and limited early-stage government/private capital make scaling up difficult, especially in high-capex sectors like drones and satellite systems

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India now has around 1,000 defence-tech start-ups actively working on dual-use technologies like autonomous systems, AI-enabled intelligence, and hyperspectral satellite payloads, backed by programmes such as iDEX and ADITI

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India increased its defence budget from ₹6.2 lakh crore to ₹6.8 lakh crore in the latest Union Budget, reflecting a consistent upward trajectory

technologies. India's defence-tech start-up ecosystem is not just innovating at speed, it's helping rewrite the national security playbook.

Major General (Retd) Ramesh Chandra Padhi, who served in the Indian Army for 36 years and is the Senior Vice President of IG Drones, says Indian start-ups have a critical role in modernising defence infrastructure. "Start-ups in India can bridge the technology gap by delivering today's technology—not yesterday's," he says.

To be sure, the Defence Research and Development Organisation (DRDO) remains the primary agency for military research and development (R&D). Padhi, however, notes that the organisation's processes are often snagged by bureaucratic red tape.

"Of course, high-end platforms like hypersonic or supersonic missiles, heavy military aircraft, advanced tanks, submarines and fifth-generation fighter jets will still need to be developed by the government or defence PSUs," he explains. These require massive capital and scale, which the private sector, as of today, does not have, he adds.

But start-ups can significantly accelerate innovation and plug operational gaps more efficiently in areas such as drones, surveillance systems, AI-based intelligence, and rapid prototyping, says Padhi.

## NEW BACKBONE

Proving the point is Pixxel, another start-up in Indian space tech, founded by BITS Pilani alumni Awais Ahmed and Kshitij Khandelwal in 2019.

Ahmed says from developing a modular 150 kg satellite capable



**In the last Budget, almost ₹1.5 lakh crore was reserved for new equipment and modernisation of the defence sector. Therefore, the total addressable market is rather large"**

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VENTURES

of integrating diverse payloads—optical, synthetic aperture radar, infrared, and hyperspectral cameras—for defence applications like missile launch detection, border surveillance, and troop-movement tracking beyond satellites, Pixxel has also secured contracts to build specialised hyperspectral and infrared cameras, strengthening India's space-based defence capabilities.

One of the critical factors behind the trend has been the role of the government. iDEX was launched with a budgetary support of ₹498.78 crore for a period of five years from 2021-22 to 2025-26 as part of a programme to provide financial support to nearly 300 start-ups.

Simultaneously, India has in